

The seam form — [FROZEN INTERNAL NOTE, NOT PUBLICATION – READY]

Dritëro M.

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Abstract

Fix the two quantized cat maps W_1, W_2 of the golden and silver monodromies at level $N = 15 = \text{tr}(A_1 A_2)$. Watch their interaction through the parity operator Par (the quantized orientation-reversal): the seam trace $(a, b) \mapsto \text{tr}(\text{Par} \cdot P_a \cdot Q_b)$, with P_a, Q_b the spectral projectors of W_1 and W_2 , lives in the cyclotomic field $\mathbb{Q}(\zeta_{60})$. Its $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ -isotypic projection — the Galois average over $\text{Gal}(\mathbb{Q}(\zeta_{60})/\mathbb{Q}(\sqrt{5}, \sqrt{-3}))$, which we call the *seam form* — is defined on a 49-point support of the (a, b) order grid and takes exactly **30 distinct values, all in $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$** . (The raw trace is non-real and degree-8 over $\mathbb{Q}$; it is the biquadratic shadow of the interaction that carries the structure below.) Fourier-transforming its four Galois channels $\{1, \sqrt{5}, \sqrt{-3}, \sqrt{-15}\}$ to the dual torus $\mathbb{Z}/20 \times \mathbb{Z}/12$, a channel vanishes only where its cyclotomic support is empty, and the joint vanishing pattern realizes exactly **four surviving fields plus the zero stratum**: the full field $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ (120 points), $\mathbb{Q}(\sqrt{5})$ (20), $\mathbb{Q}(\sqrt{-3})$ (20), the rational floor $\mathbb{Q}$ (10), and **0** — the *dark locus* — at 70 points. The one quadratic subfield that is **never** realized is $\mathbb{Q}(\sqrt{-15})$, the field of the level itself: its channel dies in every nontrivial pattern. So the selection rule is not the whole subfield lattice, but the lattice *with the $\sqrt{-15}$ node deleted and a zero stratum adjoined* — a broken lattice, and the break is the seam-level's own field. Of the 70 dark points, 54 sit at quantum closure ($[W_1^J, W_2^I] = I$). The whole architecture — the light, the dark, and the commutator trace beside them — **factors through the divisor lattice of the two operator clocks**: two functions on the thirty-six cells $(\gcd(x, 20), \gcd(y, 12))$. The proportions of light and darkness are the divisor arithmetic of the clocks that measure them. —

FROZEN (2026-07-09) as an internal note. After three adversarial re-panel rounds, this write-up still carries *framing/labelling* defects (documented in `SCRUTINY_P1P3_round3.md`); solo iteration did not converge. It is retained as an internal record only. **The verified mathematics lives in the cited B-nodes** (the seam bank B358–B367, B459, B474); do not circulate this as a paper. P4 is the only paper that cleared the gauntlet.

1 Introduction

An interaction is easy to write down and hard to see. Two operators, a product, a trace — the recipe is three symbols long, and it hides everything. This paper is about one interaction seen clearly: two quantized cat maps, watched through a single involution, and the exact proportions of light and darkness that come out. The surprise is not that structure appears — an arithmetic object will always have structure — but that the structure is *arithmetic all the way down*: the proportions of what survives and what cancels turn out to be the divisor lattice of the two clocks by which the operators keep time. Nothing is fitted. The seam form is a shadow, and the shape of the shadow is a fact about the number $15 = 3 \cdot 5$.

The two objects are the Weil / Hannay–Berry quantizations W_1, W_2 of the golden and silver monodromies $A_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, A_2 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ — the first two members of the metallic family (Paper 4) — at the level $N = 15$ that is their own product trace. The matrices A_1, A_2 are hyperbolic (Anosov); it is their *commutator* whose trace governs the interaction. The involution is Par , the quantized image of $-I$, with $\det \text{Par} = -1$: the operator through which orientation is read. The seam form is what the pair looks like through Par .

Three domains appear below, and the paper’s first job is to keep them apart, because v1 did not. (i) The *value torus*: the (a,b) grid of projector indices, on which the seam form takes its 30 values. (ii) The *dual torus*: the Fourier transform $\mathbb{Z}/20 \times \mathbb{Z}/12$ of the four channels, on which the 240 vanishing patterns and the dark locus live. (iii) The *divisor lattice*: the 36 cells ($\gcd(x,20), \gcd(y,12)$) through which the commutator trace and the pattern-tier factor. Section 2 defines the form and states its values on (i); Section 3 proves the channel-vanishing lemma; Sections 4 and 5 read the selection rule and the dark locus on (ii); Sections 6 and 7 give the spectral fingerprint and the master theorem on (iii). The honesty protocol is Paper 4’s: classical material is cited, the one deferred step is named, and every number carries a reproducer.

2 The seam form

Let $\zeta = e^{\{2\pi i/15\}}$. On $\mathbb{C}[\mathbb{Z}/15]$: the twist $D = \text{diag}(\zeta^{\{j(j-1)/2\}})$, the unitary DFT F , $W_R = (F D F)$, $W_L = D$, and $W_m = W(R^m L^m)$ — the quantized cat map of A_m at level 15. Par is the parity operator ($\text{Par } \psi)(x) = \psi(-x)$. The orders are $\text{ord}(W_1) = 20, \text{ord}(W_2) = 12$; write P_a ($a \in \mathbb{Z}/20$) and Q_b ($b \in \mathbb{Z}/12$) for the spectral projectors of W_1 and W_2 .

Definition 2.1 (the seam trace and the seam form). The *seam trace* of the pair is $\tilde{s}(a,b) = \text{tr}(\text{Par} \cdot P_a \cdot Q_b) \in \mathbb{Q}(\zeta_{60})$. The Galois group $\text{Gal}(\mathbb{Q}(\zeta_{60})/\mathbb{Q}(\sqrt{5}, \sqrt{-3})) \cong (\mathbb{Z}/2)^2$ acts on it; the **seam form** is the $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ – *isotypic* projection $s(a,b) = (1/4) \sum_{\gamma \in \text{Gal}(\mathbb{Q}(\zeta_{60})/\mathbb{Q}(\sqrt{5}, \sqrt{-3}))} \gamma \cdot \tilde{s}(a,b)$ — the average that lands in the biquadratic field. Everything below is about s .

Theorem 2.2 (values). *The raw seam trace $\tilde{s}$ is non-real at all 49 support points (a degree-8 element of $\mathbb{Q}(\zeta_{60})$); its projection s is supported on the same 49 points of the (a,b) grid (a ranges over $\{0,1,4,5,6,9,11,14,15,16,19\}$, b over $\{0,1,2,3,4,5,7,8,9,10,11\}$, with 49 of the pairs occurring), and on those 49 points s takes exactly **30 distinct values, all nonzero, all in $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$** . Its four Galois channels — the coordinates in the basis $\{1, \sqrt{5}, \sqrt{-3}, \sqrt{-15}\}$ of $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$, the isotypic pieces under $\text{Gal}(\mathbb{Q}(\sqrt{5}, \sqrt{-3})/\mathbb{Q}) \cong (\mathbb{Z}/2)^2$ — are the natural coordinates for everything below. (Their per-channel nonzero supports are 32, 45, 28, and 44 of the 49 points; the last number, 44 — the $\sqrt{-15}$ – channel support — is not the count of distinct values, a conflation corrected from v1.)*

The field $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ is the compositum of the two ends of the object: $\sqrt{5}$ from the golden (spherical, E_8) end, $\sqrt{-3}$ from the hyperbolic (E_6) end. The seam is where they meet, and its values speak both languages at once.

3 The channel-vanishing lemma

Everything downstream rests on one observation, and it is the honest heart of the paper.

Observation S5 (channel vanishing = empty support; verified, not yet proved per channel). A Galois channel of the (Fourier-transformed) seam form vanishes at a dual-torus point (x,y) if and only if the cyclotomic support of the corresponding character sum is empty. Vanishing is never an accident of magnitude; it is forced by which cyclotomic frequencies the

character can carry. Two channels of a value are therefore not free to vanish independently — their supports are governed by the same residue data, and the joint pattern is constrained.

Status. This is verified exhaustively on all 240 dual-torus points (reproducer below) and is the mechanism behind every selection rule in the paper, but a per-channel derivation from the cyclotomic support — turning the verified statement into a theorem — is **the paper's central open problem** (Section 8). It is stated here as an Observation, not with a proof mark.

Why it should hold. Two distinct roles of the Galois action must be kept apart (they are, in §8). First, the *decomposition* into four channels is Galois-isotypic: each channel is a component of the seam trace under $\text{Gal}(\mathbb{Q}(\zeta_{60})/\mathbb{Q}(\sqrt{5}, \sqrt{-3}))$, and this splitting is Galois-equivariant in the standard sense. Second, the *vanishing* of a channel is a separate question — and, as §8 records, it is **not** produced by the standard signed-permutation Galois action on modular data (which never sends a nonzero entry to zero). What we observe is that a channel, being a sum over a cyclotomic residue class, vanishes exactly when that class carries no surviving frequency; turning this "should" into a per-channel theorem — a *support/vanishing* criterion, not an equivariance statement — is what remains.

The Observation is why the dark locus is a *structural* fact and not a table. Light is the generic case; a point goes dark only when arithmetic forbids every channel at once, and forbidding is a congruence condition, not a coincidence.

4 The selection rule: a broken subfield lattice

Theorem 4.1 (five patterns; four fields and the dark stratum). *Exactly five vanishing patterns occur across the 240 dual-torus points. Written as the set of channels that vanish, with the surviving field and the count:*

pattern	channels that vanish	surviving field	count
free	none	$\mathbb{Q}(\sqrt{5}, \sqrt{-3})$	120
rs	$\sqrt{-3}, \sqrt{-15}$	$\mathbb{Q}(\sqrt{5})$	20
qs	$\sqrt{5}, \sqrt{-15}$	$\mathbb{Q}(\sqrt{-3})$	20
qrs	$\sqrt{5}, \sqrt{-3}, \sqrt{-15}$	$\mathbb{Q}$ (rational)	10
dark	all four	0	70

These realize four of the five nodes of the subfield lattice of $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ — the whole field, the two quadratic subfields $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, and the rational floor — **together with the zero stratum**. The fifth lattice node, $\mathbb{Q}(\sqrt{-15})$, is **never realized**: the pattern in which only the $\sqrt{-15}$ channel survives ($\sqrt{5}, \sqrt{-3}$ dead, $\sqrt{-15}$ alive) does not occur at any of the 240 points. The $\sqrt{-15}$ channel instead *dies in every nontrivial pattern* — it is the first channel to go and the one that never survives alone. So the selection rule is the subfield lattice **with the $\mathbb{Q}(\sqrt{-15})$ node deleted and a zero stratum adjoined**: a broken lattice, and the break is exactly the field of the level, $\mathbb{Q}(\sqrt{-15})$.

(Correction from v1, recorded for the reader: v1 called the five patterns "precisely the five nodes of the subfield lattice, including $\mathbb{Q}(\sqrt{-15})$." That is wrong twice over — $\mathbb{Q}(\sqrt{-15})$ is never a surviving field, and the zero stratum, present at 70 points, is not a subfield-lattice node. The broken-lattice-plus-zero statement above is the corrected and, we think, more pointed one.)

5 The dark locus

Theorem 5.1 (darkness concentrates at closure). *Of the seventy dark points, **fifty-four sit at quantum closure** — the addresses (j,l) where $[W_1^j, W_2^l] = I$. Darkness is not scattered; it pools where the two operators commute.*

This is the paper’s cleanest image. Where the interaction vanishes entirely — where the object, watched through Par, shows nothing — is overwhelmingly where the two motions have come into phase and their commutator has died. The light is the interference; the dark is the silence after the two clocks agree. Sixteen of the seventy dark points lie *off* closure, and those sixteen are the paper’s one genuinely open residue (Section 8).

6 The spectral fingerprint and the cornerstone

Theorem 6.1 (spectral invariants). *The seam operator carries the framing fingerprint $\sigma_1 = \sigma_2 = 1/24$ — the central-charge signature $c/24$ of the quantization, independent of the pair.*

Theorem 6.2 (the cornerstone). *The distinguished quadratic subfield of the seam is $\mathbb{Q}(\sqrt{-15})$ — the field of the level itself. It is the first channel to vanish and the node the reduction is set against: the one quadratic subfield that never survives, so that the surviving fields form the lattice around it.*

Theorem 6.3 (the exchange identity). *The half-monodromy exchange $\sigma : W_1 \leftrightarrow W_2$ (the Dehn-filling pair, the Gieseking deck action) acts on the seam values by an explicit involution; the seam form is symmetric under it up to the orientation sign carried by Par.*

7 The master theorem: two functions on the divisor lattice

Here the spine’s promise is paid.

Theorem 7.1 (the divisor-lattice factorization). *Both maps on the order torus $\mathbb{Z}/20 \times \mathbb{Z}/12$ — the commutator trace $\kappa_q(j,l) = \text{tr}[W_1^j, W_2^l]$ and the tier of the seam form (its vanishing pattern) — factor through the divisor pair $(\gcd(x,20), \gcd(y,12))$. Each is single-valued on the thirty-six divisor cells. The commutator trace factors as $\kappa_q = \varepsilon(jl) \cdot \chi_5$, where $\varepsilon \in \{3, -1\}$ is the quaternionic parity (from the mod-3 image Q_8) and $\chi_5 \in \{5, 1\}$ is the mod-5 closure bit. The seam tier factors by the same even-function mechanism (Cohen/McCarthy (r,s) -even functions; Serre’s $\mathbb{Q}$ – class principle) applied channel-wise.*

Theorem 7.2 (the magnitude law). $|\kappa_q(j,l)|^2 = \#\text{Fix}([A_1^j, A_2^l])$ acting on $(\mathbb{Z}/15)^2$ — Howe’s formula. Verified on the 5×5 principal window (all 25 cells) at level 15; the general 36-cell statement is asserted on that verified window and the even-function factorization, with the full-torus check flagged as remaining.

The master table — thirty-six cells, each carrying its κ_q value and its tier — is the paper’s final object, printed in full in the appendix. It is the answer to the spine: the light and the dark of the interaction, and the commutator that measures how far the operators are from commuting, are the *same* two functions, and both are read off the greatest common divisors of the exponents with the two clock orders. The proportions of light and darkness are the divisor arithmetic of the clocks.

That the factorization mechanism is classical (even functions; $\mathbb{Q}$ – *classes*) is stated plainly: what is ours is the specific $N = 15$ seam, its broken-lattice selection rule (four fields plus zero,

with $\mathbb{Q}(\sqrt{-15})$ absent), the *dark – locus/closure* concentration, and the joint reading of κ_q and the tier on one lattice.

8 What we could not locate, and what remains open

The classical shelf: Weil-representation character values (Howe; Thomas; and, at odd composite level, the explicit formulas of Ladisch and of Strömberg), the *Hecke/CRT* factorization from the real-quadratic eigenvalue field (Kurlberg–Rudnick), the general Galois symmetry of modular data (Coste–Gannon; Bantay; Dong–Lin–Ng), the even-function class (Cohen; McCarthy), and the $\mathbb{Q}$ – *class* principle (Serre). Against that shelf, our search — an adversarial prior-art assessment run after this paper’s panel — did not locate: the explicit 30-value seam form at any composite level; the broken-lattice selection rule with $\mathbb{Q}(\sqrt{-15})$ systematically absent; the *dark – locus/closure* concentration (54 of 70); or the joint (κ_q, tier) divisor-lattice factorization. The verdict is **APPEARS-NOVEL / NEEDS-SPECIALIST**, and one structural observation sharpens it: the general Galois symmetry of modular data (Dong–Lin–Ng) acts by a *signed permutation* $G_\sigma = \sigma(s)s^{-1}$, which can never send a nonzero matrix entry to zero — so the systematic vanishing of the $\sqrt{-15}$ channel is **not** an instance of the standard Galois-equivariance mechanism, and the selection rule is a distinct phenomenon. This is a negative-existence verdict over the canonical references; a specialist in Weil-representation Galois theory should confirm no *CRT/Galois – descent* argument in less-indexed literature explains the $\sqrt{-15}$ absence.

Open: (i) the **cyclotomic-support step** (Observation S5) — deriving each cell’s tier from its support is verified on all 240 dual-torus points but not yet proved per channel; it is the paper’s central open problem. (ii) The **sixteen off-closure dark points** — the dark locus is not exhausted by the closure set, and the residue is unexplained. (iii) The **seam at other levels** — whether the broken-lattice selection rule survives at levels other than $3 \cdot 5$, and how it degrades when a prime goes trivial. (iv) The **full-torus S10 check** — Howe’s magnitude law is verified on the 5×5 principal window; the remaining cells are asserted via the factorization, not yet exhaustively checked.

9 Appendix A: the master table

(The 36-cell table $(\gcd(x,20), \gcd(y,12)) \mapsto (\kappa_q, \text{tier})$, printed in full as in Paper 4 §4.3; reproducer `B474_tier_commutator_law/cross_table.py`, exact at $p = 61, \mathbb{Q}(\zeta_{60})$.)

10 Appendix B: reproducibility

Per-claim script map: S1–S4/S11 (the seam bank B358–B367, B449; the 30 – *value/49 – support* count, the per-channel supports $\{32,45,28,44\}$, and the $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ – *projection* definition recomputed at revision via `B358_seam_certification + B367_value_map` — the raw trace’s *degree – 8/non – real* status confirmed at all 49 points); S5–S6 (`B459_klein_four_verification/verify_patterns.py` — the five patterns, counts 120/20/20/10/70, the two laws, and the never-realized $\mathbb{Q}(\sqrt{-15})$ node, all reproduced at revision); **S7’s dark – locus/closure concentration (54 of 70) is NOT produced by `verify_patterns.py`** (which computes only the patterns and their counts); it is the intersection of the 70-point dark set with the commutator-closure locus $[W_1^j, W_2^l] = I$, and requires its own reproducer — flagged here as a step to be committed with the paper, not yet part of the reproduced set. S8/S12 (`br1_br2.py`, B469); S9/S10 (`cross_table.py`, `kq_verify.py`, B474/B472). Each carries a lock re-run at draft. The $v1 \rightarrow v4$ corrections (30 distinct values not 44; the broken lattice with $\mathbb{Q}(\sqrt{-15})$ absent and the zero stratum present; the value-torus

/ dual-torus separation; Observation S5 demoted from proved to the central open problem; and — v4 — the seam form redefined as the $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$ — *projection*, since the raw trace is *degree* — 8/*non* — *real*) are recorded in the corrections ledger. The earlier float miscount of the dark set as 53 rather than the exact 70 is also recorded there.

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